

General Chemistry (Chem 1006)

Chapter 4

Stoichiometry of Chemical Reactions

4.1. Writing and Balancing Chemical Equations

4.1.1. Writing Chemical Equations

- **Element symbols** represent individual atoms and
- **Chemical formulas** represent compounds

- The identities and the relative quantities of substances undergoing a chemical change are represented by writing and balanced **chemical equation.**
- **Chemical equation** uses element symbols and chemical formulas.

✓ The physical states of reactants and products are indicated with a parenthetical abbreviation (such as: s, l, g, *aq*) following the formulas.

Example: The chemical equation for the reaction between methane and oxygen to produce carbon dioxide and water is:

$$CH_4 + 2O_2 \longrightarrow CO_2 + 2H_2O$$

- **Reactants:** The substances undergoing reaction
  - ➢ Their formulas are placed on the left side of the equation.
- **Products:** The substances generated by the reaction
  - ➢ Their formulas are placed on the right side of the equation.
- **Plus signs** (+) separate individual reactant and product formulas
- **An arrow** (⟶) separates the reactants and the products
- **Coefficients:** The relative numbers of reactants and products
  - ➢ They are represented by numbers written to the left of each formula
  - ➢ A coefficient of 1 is typically omitted
- In a chemical equation, the smallest possible whole-number coefficients are used

✓ The coefficients show the ratios of the species.

Example: Methane and oxygen react to yield carbon dioxide and water in a 1:2:1:2 ratio, respectively.

- One methane molecule and two oxygen molecules react to yield one carbon dioxide molecule and *two* water molecules.
- One mole of methane molecules and 2 moles of oxygen molecules react to yield 1 mol*e* of carbon dioxide molecules and *2* moles of water molecules.
- Necessary conditions for a reaction are sometimes designated by writing a word or symbol above or below the equation's arrow.

Example: The Δ represents the requirement of heat for the reaction:

$$CaCO_3(s) \xrightarrow{\Delta} CaO(s) + CO_2(g)$$

4.1.2. Balancing Chemical Equations

- **Balanced equation:** A chemical equation that has equal numbers of atoms for each element on the reactant and product sides. This is a requirement the equation must satisfy to be consistent with the law of conservation of matter
- Note that the number of atoms for a given element is calculated by multiplying the coefficient of any formula containing that element by the element's subscript in the formula.

Example: In the equation: $CH_4 + 2O_2 \longrightarrow CO_2 + 2H_2O$

The numbers of carbon, oxygen and hydrogen atoms on each side of the equation are 1, 4 and 4, respectively.

➢ Therefore the equation is balanced.

In the reaction between N_2 and O_2 to produce N_2O_5, the equation

$$N_2 + O_2 \longrightarrow N_2O_5$$ is unbalanced

$$2N_2 + 5O_2 \longrightarrow 2N_2O_5$$ is balanced

- A balanced chemical equation may be derived by an approach known as "balancing by inspection"

Example: Balance the following reaction.



First balance oxygen and then iron.

Answer: 

4.1.3. Equations for Ionic Reactions

- Consider a reaction between ionic compounds taking place in an aqueous solution.

Example:

- When aqueous solutions of $CaCl_2$ and $AgNO_3$ are mixed, a reaction takes place producing aqueous $Ca(NO_3)_2$ and solid AgCl.



- The equation is called a **molecular equation** because it doesn't explicitly represent the ionic species that are present in solution
- Ionic compounds dissolved in water are more realistically represented as dissociated ions
- An equation that represents all dissolved ions is called a **complete ionic equation**

- **Molecular and Ionic Equations**

Example: Write balanced molecular, complete ionic, and net ionic equations for the following reaction.

$CO_2(g) + NaOH(aq) \longrightarrow Na_2CO_3(aq) + H_2O(l)$

Solution

Balanced molecular equation:

$CO_2(g) + 2NaOH(aq) \longrightarrow Na_2CO_3(aq) + H_2O(l)$

Balanced complete ionic equation

$CO_2(g) + 2Na^+ + 2OH^-(aq) \longrightarrow 2Na^+_{(aq)} + CO_3^{2-}(aq) + H_2O(l)$

Balanced net ionic equation: Remove the spectator ion (Na^+)

$CO_2(g) + 2OH^-(aq) \longrightarrow CO_3^{2-}(aq) + H_2O(l)$

4.2. Classification of Chemical Reactions

4.2.1. Precipitation reactions and solubility rules

Precipitation reaction: A reaction in which dissolved substances react to form one or more solid products.

- Many reactions of this type involve the exchange of ions between ionic compounds in aqueous solution
- They are sometimes referred to as double displacement, double replacement, or metathesis reactions.

They are common

- ✓ in industry for production of commodities
- ✓ in nature

Example: They are responsible for the formation of kidney stones

Example:

$$2KI(aq) + Pb(NO_3)_2(aq) \longrightarrow PbI_2(s) + 2KNO_3(aq)$$

Net ionic equation:

$$Pb^{2+}(aq) + 2I^-(aq) \longrightarrow PbI_2(s)$$

Solubility: The maximum concentration of a substance that can be dissolved in water, or any solvent under specified conditions such as temperature.

- Substances with relatively large solubilities are said to be **soluble**.
- ➢ A substance will **precipitate** in the solution if its concentration exceeds its solubility.
- Substances with relatively low solubilities are said to be **insoluble**
- ➢ These are the substances that readily precipitate from solutions.

Soluble compounds contain	Exceptions
group 1 cations (Li^+, Na^+, K^+, Rb^+ Cs^+) ammonium ion (NH_4^+)	
halide ions (Cl^-, Br^-, and I^-)	halides of Ag^+, Hg_2^{2+}, Pb^{2+}
acetate ($C_2H_3O_2^-$), bicarbonate (HCO_3^-), nitrate (NO_3^-), chlorate (ClO_3^-), sulfate (SO_4^{2-})	sulfates of Ag^+, Ba^{2+}, Ca^{2+}, Hg_2^{2+}, Pb^{2+}, and Sr^{2+}

Insoluble compounds contain	Exceptions
carbonate (CO_3^{2-}), chromate (CrO_3^{2-}), phosphate (PO_4^{3-}), sulfide (S^{2-}),	compounds of these anions with group 1 cations and NH_4^+,
hydroxide (OH^-)	hydroxides of group 1 cations and Ba^{2+}

Examples of Precipitation Reactions:

$2KI(aq) + Pb(NO_3)_2(aq) \longrightarrow PbI_2(s) + 2KNO_3(aq)$ molecular equation

$Pb^{2+}(aq) + 2I^-(aq) \longrightarrow PbI_2(s)$ Net ionic equation

$NaF(aq) + AgNO_3(aq) \longrightarrow AgF(s) + NaNO_3(aq)$ molecular equation

$Ag^+(aq) + F^-(aq) \longrightarrow AgF(s)$ net ionic equation

Exercise (Class work):

Predict whether a precipitation reaction will occur when aqueous solutions of the following ionic compounds are mixed together.

a) potassium sulfate and barium nitrate

b) lithium chloride and silver acetate

Solution:

a)

- $BaSO_4$ is insoluble
- Precipitation reaction is expected

$$K_2SO_4\,(aq) + Ba(NO_3)_2\,(aq) \rightarrow KNO_3\,(aq) + BaSO_4(s)$$

Net ionic equation:

$$Ba^{2+} + SO_4^{2-} \rightarrow BaSO_4(s)$$

b)

- AgCl is insoluble
- Precipitation reaction is expected

$$LiCl\,(aq) + AgC_2H_3O_2\,(aq) \rightarrow LiC_2H_3O_2\,(aq) + AgCl\,(s)$$

Net ionic equation:

$$Ag^+(aq) + Cl^-(aq) \longrightarrow AgCl(s)$$

4.2.2. Acid-base reactions

- An **acid-base reaction** is one in which a hydrogen ion, H^+, is transferred from one chemical species to another.
- Such reactions are important to numerous natural and technological processes, ranging from the
  - ✓ chemical transformations that take place within cells
  - ✓ industrial-scale production of fertilizers, pharmaceuticals, and other substances.
- Common types of acid-base reactions that take place in aqueous solutions are considered.

Acid: A substance that will dissolve in water to yield hydronium ions, H_3O^+.

Example

$HCl(aq) + H_2O(aq) \longrightarrow Cl^-(aq) + H_3O^+(aq)$

- The process represented by the equation confirms that hydrogen chloride is an acid.

➢ When dissolved in water, H_3O^+ ions are produced by a chemical reaction in which H^+ ions are transferred from HCl molecules to H_2O molecules

✓ Every HCl molecule that dissolves in water will undergo this reaction.

✓ The reaction is 100% efficient

Strong acids: Acids that are completely or nearly **100% ionized** in their aqueous solutions

Examples of strong acids: HCl, HNO_3, H_2SO_4, HBr, HI, $HClO_4$, $HClO_3$

Weak acids: Acids that are only **partially ionized** in their solutions

- Do not completely dissociate into their ions in water
- They partially react with water
- Relatively small amount of hydronium ions are formed
- Large majority of dissolved molecules remain in their original form

Example

$$CH_3COOH(aq) + H_2O(l) \rightleftharpoons CH_3COO^-(aq) + H_3O^+(aq)$$

✓ About 1% of acetic acid molecules are present in the ionized form, $CH_3CO_2^-$

Examples of weak acids:

$HO_2C_2O_2H$ - oxalic acid

HF – hydrofluoric acid

C_6H_5COOH -benzoic acid

CH_3COOH - acetic acid (Ethanoic acid)

HCOOH - formic acid (methanoic acid)

Conjugate acids of weak bases (e.g. NH_4^+), H_2O

Base: A substance that will dissolve in water to yield hydroxide ions, OH^-.

- The most common bases are ionic compounds composed of alkali or alkaline earth metal cations (groups 1 and 2) combined with the hydroxide ion

Examples: NaOH, $Ca(OH)_2$

➢ When dissolve in water, hydroxide ions are released directly into the solution.

Strong bases: Bases that completely dissociate in water.

$$NaOH(s) \longrightarrow Na^+(aq) + OH^-(aq)$$

- The equation confirms that sodium hydroxide is a base.

➢ When dissolved in water, NaOH dissociates to yield Na^+ and OH^- ions.

➢ The dissociation process is essentially complete (100% efficient)

Weak bases: A base that upon dissolution in water, does not dissociate completely.

- ➢ the resulting aqueous solution contains only a small proportion of hydroxide ions
- ▪ They produce hydroxide ions when dissolved by chemically reacting with water molecules.

Example:

$$NH_3(aq) + H_2O(l) \rightleftharpoons NH_4^+(aq) + OH^-(aq)$$

- ✓ It is an acid-base reaction
- ✓ Only about 1% of the dissolved ammonia is present as NH_4^+ ions.

Examples of weak bases:

- ✓ Ammonia (NH_3), ammonium hydroxide (NH_4OH), trimethyl ammonia ($N(CH_3)_3$), H_2O, conjugate bases of weak acids (e.g. $HCOO^-$)

Neutralization reaction: It is a specific type of acid-base reaction in which the reactants are an acid and a base;

✓ The products are often a **salt** and water

acid + base ⟶ salt + water

Example:

The antacid milk of magnesia (aqueous suspension of solid $Mg(OH)_2$) when ingested to ease symptoms associated with excess stomach acid (HCl), undergoes the following neutralization reaction:

$$Mg(OH)_2(s) + 2HCl(aq) \longrightarrow MgCl_2(aq) + 2H_2O(l)$$

Exercise: Write balanced chemical equations for the acid-base reactions:

a) hydrogen hypochlorite(weak acid) reacts with water

b) a solution of barium hydroxide is neutralized with a solution of nitric acid

Solution

a) $HOCl(aq) + H_2O(l) \rightleftharpoons OCl^-(aq) + H_3O^+(aq)$

b) $Ba(OH)_2(aq) + 2HNO_3(aq) \longrightarrow Ba(NO_3)_2(aq) + 2H_2O(l)$

Exercise: Write the net ionic equation representing the neutralization of any strong acid with an ionic hydroxide.

Solution:

Consider the reaction between HCl(aq)and NaOH(aq)

Molecular equation:

$HCl\ (aq) + NaOH(aq) \rightarrow NaCl(aq) + H_2O$

Ionic equation:

$H_3O^+ + Cl^-\ (aq) + Na^+\ (aq) + OH^-\ (aq) \rightarrow Na^+\ (aq) + Cl^-\ (aq) + 2H_2O$

Net ionic equation:

$H_3O^+\ (aq) + OH^-\ (aq) \rightarrow 2H_2O$ – It hold for similar reactions

4.2.3. Oxidation-Reduction Reactions (Redox reaction):

- A reaction characterized by the transfer of electrons between chemical species

Oxidation: loss of electrons

- The species that **losses electrons** is said to be oxidized

Reduction: gain of electrons

- The species that **gains electrons** is said to be reduced

- **Reducing agent (reductant):** A species that is oxidized
- **Oxidizing agent (oxidant):** A species that is reduced

Consider the reaction between sodium and chlorine to yield sodium chloride:

$$2Na(s) + Cl_2(g) \longrightarrow 2NaCl(s)$$

- The process can be viewed with regard to the fate of each reactant in the form of an equation called a **half-reaction**

  Oxidation Half reaction: $2Na(s) \longrightarrow 2Na^+(s) + 2e^-$

  Reduction Half reaction: $Cl_2(g) + 2e^- \longrightarrow 2Cl^-(s)$

- These equations show that Na atoms lose electrons while Cl atoms (in the Cl_2 molecule) gain electrons
- NaCl is an ionic compound

- Some redox processes, however, do not involve the transfer of electrons.

➢ Consider the following reaction.

$$H_2(g) + Cl_2(g) \longrightarrow 2HCl(g)$$

✓ The product of this reaction is a covalent compound,

✓ Transfer of electrons in the explicit sense is not involved

Oxidation number (**oxidation state**): Oxidation number of an element in a compound is the charge its atoms would possess *if the compound was ionic*.

Guidelines for assigning oxidation numbers to each element in a molecule or ion:

1. The oxidation number of an atom in an elemental substance is zero
2. The oxidation number of a monatomic ion is equal to the ion's charge.
3. Oxidation numbers for common nonmetals:

Hydrogen:

✓ +1 when combined with nonmetals

✓ −1 when combined with metals

Oxygen:

−2 in most compounds −1 in peroxides (O_2^{2-})

rarely −½ in superoxides (O_2^-) positive values with F (values vary)

Halogens:

−1 for F always

−1 for other halogens except when combined with oxygen or other halogens (+ve oxidation numbers in these cases, varying values)

4. The sum of oxidation numbers for all atoms in a molecule or polyatomic ion is equal to the charge on the molecule or ion.

Note:

The proper convention for

- ✓ reporting charge is to write the number first, followed by the sign (e.g., 2+)
- ✓ oxidation number is written with the reversed sequence, sign followed by number (e.g., +2).

Examples:

Determine the oxidation number of each element in the following compounds.

a) NaH b) H_2SO_4

Solution:

a) Following guideline 3, the oxidation number for H is -1

Following guideline 4,

charge on NaH = 0 = oxidation number of Na + oxidation number of H

- oxidation number of Na = 0 – (-1) = +1

b) For ionic compounds, it's convenient to assign oxidation numbers for the cation and anion separately.

According to guideline 3, the oxidation number for hydrogen is +1.

Following guideline 4,

Oxidation number of H_2SO_4 = 0 = 2 x oxidation number of H + oxidation number of SO_4

oxidation number of SO_4 = 0 – (2 x +1) = -2

Following guideline 3, oxidation number of O = -2.

oxidation number of SO_4 = -2 = oxidation number of S + (4 x -2)

oxidation number of S = -2 + 8 = +6

Using the oxidation number concept:

Oxidation-reduction(redox) reactions: Reactions in which one or more elements involved undergo a change in oxidation number.

Accordingly,

- ✓ **Oxidation:** An increase in oxidation number
- ✓ **Reduction:** A decrease in oxidation number

In the reaction:

$2Na(s) + Cl_2(g) \longrightarrow 2NaCl(s)$

Na is oxidized

- ➢ Its oxidation number increases from 0 in Na to +1 in NaCl

  Cl is reduced
- ➢ Its oxidation number decreases from 0 in Cl_2 to −1 in NaCl

Subclasses of redox reactions

1. Combustion reactions: The reductant (a *fuel*) and oxidant (often, O_2) react vigorously and produce significant amounts of heat, and often light, in the form of a flame.

Example:

$$10Al(s) + 6NH_4ClO_4(s) \longrightarrow 4Al_2O_3(s) + 2AlCl_3(s) + 12H_2O(g) + 3N_2(g)$$

2. Single-displacement (replacement) reactions:

Redox reactions in which an ion in solution is displaced (or replaced) via the oxidation of a metallic element.

Example

$$Zn(s) + 2HCl(aq) \longrightarrow ZnCl_2(aq) + H_2(g)$$

$$Cu(s) + 2AgNO_3(aq) \longrightarrow Cu(NO_3)_2(aq) + 2Ag(s)$$

3. Disproportionation reactions: The same substance functions as an oxidant and a reductant.

Example:

$$2H_2O_2(aq) \longrightarrow 2H_2O(l) + O_2(g)$$

- This is a redox reaction
- Oxygen is oxidized,
- Its oxidation number increasing from −1 in $H_2O_2(aq)$ to 0 in $O_2(g)$.
- Oxygen is reduced, its oxidation number decreasing from −1 in $H_2O_2(aq)$ to −2 in $H_2O(l)$.

Examples

Identify which equations represent redox reactions, providing a name for the reaction if appropriate. For those reactions identified as redox, name the oxidant and reductant.

a) $ZnCO_3(s) \longrightarrow ZnO(s) + CO_2(g)$

b) $C_2H_4(g) + 3O_2(g) \longrightarrow 2CO_2(g) + 2H_2O(l)$

Solution:

a) The oxidation number of Zn, C and O on both sides of the equation are +2, +4 and -2, respectively

- Oxidation numbers remain unchanged for all elements.

➢ The reaction is not a redox reaction

b) This is a redox reaction

➢ It is combustion reaction.

➢ Carbon is oxidized

✓ Its oxidation number increasing from −2 in C_2H_4 to +4 in CO_2

➢ The reducing agent (fuel) is C_2H_4

➢ Oxygen is reduced

✓ Its oxidation number decreasing from 0 in O_2 to −2 in H_2O

✓ The oxidizing agent is O_2

Balancing Redox Reactions via the Half-Reaction Method

- Redox reactions that take place in aqueous media often **involve water, hydronium ions, and hydroxide ions as reactants or products**.

➢ Although these species are not oxidized or reduced, they do participate in chemical change

Steps for balancing redox reactions

1. Write the two half-reactions representing the redox process.
2. Balance all elements except oxygen and hydrogen.
3. Balance oxygen atoms by adding H_2O molecules.
4. Balance hydrogen atoms by adding H^+ ions.
5. Balance charge by adding electrons.
6. Multiply each half-reaction's coefficients by the smallest possible integers to yield equal numbers of electrons in each.
7. Add the balanced half-reactions together and simplify by removing species that appear on both sides of the equation.

For reactions **occurring in basic media**, carry out these additional steps:

7. Add OH^- ions to both sides of each equation in numbers equal to the number of H^+ ions.
8. On the side of the equation containing both H^+ and OH^- ions, combine these ions to yield water molecules.
9. Add the two equations and simplify the equation by removing any redundant species.

Finally, check to see that both the number of atoms and the total charges are balanced.

Example 1: Balance the following reaction that takes place in acidic solution.

$Cr_2O_7^{2-} + Fe^{2+} \longrightarrow Cr^{3+} + Fe^{3+}$

Solution:

Step 1. $Fe^{2+} \longrightarrow Fe^{3+}$ $\quad Cr_2O_7^{2-} \longrightarrow Cr^{3+}$

Step 2. $Fe^{2+} \longrightarrow Fe^{3+}$ $\quad Cr_2O_7^{2-} \longrightarrow 2Cr^{3+}$

Step3. $Fe^{2+} \longrightarrow Fe^{3+}$ $\quad Cr_2O_7^{2-} \longrightarrow 2Cr^{3+} + 7H_2O$

Step 4. $Fe^{2+} \longrightarrow Fe^{3+}$ $\quad Cr_2O_7^{2-} + 14H^+ \longrightarrow 2Cr^{3+} + 7H_2O$

Step 5. $Fe^{2+} \longrightarrow Fe^{3+} + e^-$ $\quad Cr_2O_7^{2-} + 14H^+ + 6e^- \longrightarrow 2Cr^{3+} + 7H_2O$

Step 6 . $6(Fe^{2+} \longrightarrow Fe^{3+} + e^-)$ $\quad 1(Cr_2O_7^{2-} + 14H^+ + 6e^- \longrightarrow 2Cr^{3+} + 7H_2O)$

$6Fe^{2+} \longrightarrow 6Fe^{3+} + 6e^-$ $\quad Cr_2O_7^{2-} + 14H^+ + 6e^- \longrightarrow 2Cr^{3+} + 7H_2O$

Step 7. $6Fe^{2+} + Cr_2O_7^{2-} + 14H^+ + 6e^- \longrightarrow 6Fe^{3+} + 6e^- + 2Cr^{3+} + 7H_2O$

$$\mathbf{6Fe^{2+} + Cr_2O_7^{2-} + 14H^{+-} \longrightarrow 6Fe^{3+} + 2Cr^{3+} + 7H_2O}$$

Example 2: Balance the following reaction that takes place in basic media

$$MnO_4^-(aq) + NO_2^-(aq) \longrightarrow MnO_2(s) + NO_3^-(aq)$$

Solution:

Step 1. $MnO_4^- \longrightarrow MnO_2$ $\qquad$ $NO_2^- \longrightarrow NO_3^-$

Step 2. - $\qquad$ -

Step3. $MnO_4^- \longrightarrow MnO_2 + 2H_2O$ $\qquad$ $NO_2^- + H_2O \longrightarrow NO_3^-$

Step 4. $MnO_4^- + 4H^+ \longrightarrow MnO_2 + 2H_2O$ $\qquad$ $NO_2^- + H_2O \longrightarrow NO_3^- + 2H^+$

Step 5. $MnO_4^- + 4H^+ + 3e^- \longrightarrow MnO_2 + 2H_2O$ $\qquad$ $NO_2^- + H_2O \longrightarrow NO_3^- + 2H^+ + 2e^-$

Step 6. $2(MnO_4^- + 4H^+ + 3e^- \longrightarrow MnO_2 + 2H_2O)$ $\qquad$ $3(NO_2^- + H_2O \longrightarrow NO_3^- + 2H^+ + 2e^-)$

$2MnO_4^- + 8H^+ + 6e^- \longrightarrow 2MnO_2 + 4H_2O$ $\qquad$ $3NO_2^- + 3H_2O \longrightarrow 3\,NO_3^- + 6H^+ + 6e^-$

Step 7. $2MnO_4^- + 8H^+ + 8OH^- + 6e^- \longrightarrow 2MnO_2 + 4H_2O + 8OH^-$

$3NO_2^- + 3H_2O + 6OH^- \longrightarrow 3NO_3^- + 6H^+ + 6OH^- + 6e^-$

Step 8. $2MnO_4^- + 8H_2O + 6e^- \longrightarrow 2MnO_2 + 4H_2O + 8OH^-$

$3NO_2^- + 3H_2O + 6OH^- \longrightarrow 3\,NO_3^- + 6H_2O + 6e^-$

Step 9. $\mathbf{2MnO_4^- + 3NO_2^- + H_2O \longrightarrow 2MnO_2 + 3\,NO_3^- + 2OH^-}$

Class work

Balance the following chemical equation in acidic solution

$ClO_3^- + SO_2 \rightarrow SO_4^{2-} + Cl^-$

Solution:

Split into unbalanced half-reactions:

$ClO_3^- \rightarrow Cl^-$

$SO_2 \rightarrow SO_4^{2-}$

Balance the half-reactions:

$6e^- + 6H^+ + ClO_3^- \rightarrow Cl^- + 3H_2O$

$2H_2O + SO_2 \rightarrow SO_4^{2-} + 4H^+ + 2e^-$

Make the number of electrons equal:

$6e^- + 6H^+ + ClO_3^- \rightarrow Cl^- + 3H_2O$

$6H_2O + 3SO_2 \rightarrow 3SO_4^{2-} + 12H^+ + 6e^-$ multiplied through by a factor of 3

Add the two half-reactions for the final answer:

$ClO_3^- + 3H_2O + 3SO_2 \rightarrow 3SO_4^{2-} + Cl^- + 6H^+$

Example: Balance the equation **$Au + NaCN + O_2 + H_2O \rightarrow NaAu(CN)_2 + NaOH$** in basic medium

Solution:

Net-ionic:

$$Au + CN^- + O_2 \rightarrow Au(CN)_2^- + OH^-$$

Half-reactions:

$$Au + CN^- \rightarrow Au(CN)_2^-$$

$$O_2 \rightarrow OH^-$$

Balance:

$$Au + 2CN^- \rightarrow Au(CN)_2^- + e^-$$

$$4e^- + 2H^+ + O_2 \rightarrow 2OH^-$$

Equalize electrons:

$$4Au + 8CN^- \rightarrow 4Au(CN)_2^- + 4e^-$$

$$4e^- + 2H^+ + O_2 \rightarrow 2OH^-$$

Add:

$$4Au + 8CN^- + 2H^+ + O_2 \rightarrow 4Au(CN)_2^- + 2OH^-$$

Only the electrons cancel.

Convert to basic solution:

$$4Au + 8CN^- + 2H_2O + O_2 \rightarrow 4Au(CN)_2^- + 4OH^-$$

4.3. Reaction Stoichiometry

Stoichiometry: The study of quantitative relationships between the amounts of substances consumed and produced by a reaction.

Coefficients in a balanced chemical equation provide the relative numbers of these chemical species

Example 1: In the reaction: **$H_2(g) + Cl_2(g) \longrightarrow 2HCl(g)$**

- The balanced equation shows the hydrogen and chlorine react to produce hydrogen chloride in a 1:1:2 stoichiometric ratio, respectively.

Example 2: How many moles of $Ca(OH)_2$ are required to react with 1.36 mol of H_3PO_4 to produce $Ca_3(PO_4)_2$ according to the equation: **$3Ca(OH)_2 + 2H_3PO_4 \longrightarrow Ca_3(PO_4)_2 + 6H_2O$**?

Answer: $\text{mol of } (\text{CaOH})_2 = \frac{3mol}{2\ mol} \times 1.36\ mol = 2.04\ mol$

Example 3: What mass of sodium hydroxide, NaOH, would be required to produce 16 g of the antacid milk of magnesia [magnesium hydroxide, $Mg(OH)_2$] by the following reaction?

$MgCl_2(aq) + 2NaOH(aq) \longrightarrow Mg(OH)_2(s) + 2NaCl(aq)$

(molar masses: NaOH = 40 g, $Mg(OH)_2$ =58.3 g)

Answer: $\text{mass of NaOH} = \frac{2\times 40\ g}{58.3\ g} \times 16\ g = 22\ g$

4.4. Reaction Yield

4.4.1. Limiting Reactant

Limiting reactant: The reactant provided in smaller amount than the stoichiometric amount

Excess reactant: The reactant provided in larger amount than stoichiometric amount

N.B.: The amount of the product formed depends on the amount of the limiting reactant.

Example: Consider the reaction:

$$N_{2(g)} + 3H_{2(g)} \rightarrow 2NH_{3(g)}$$

The stoichiometric ratio for N_2 , H_2 and NH_3 is 1:3:2, respectively.

Imagine combining 3 moles of N_2 and 6moles of H_2.

From the balanced equation, for 3 moles of N_2 9 moles of H_2 are required.

But, there are only 6 moles.

Therefore,

- ➢ The excess reactant is N_2
- ✓ It is in excess by 3 mole
- ➢ The limiting reactant is H_2
- ✓ The amount of NH_3 produced depends on the amount of H_2
- ✓ The amount of NH_3 that can be produced is 4 moles

4.4.2. Percent Yield

Theoretical yield: The amount of product that *may be* produced by a reaction under specified conditions, as calculated per the stoichiometry of an appropriate balanced chemical equation

Actual yield: The amount of product obtained in practice

- It is often less than the theoretical yield possible reasons:
  - ✓ Inefficient reactions because of side reaction
  - ✓ Incomplete reactions
  - ✓ Difficult to collect the products without some loss

Percent yield: The extent to which a reaction's theoretical yield is achieved

It is expressed as:

$$\boldsymbol{percentage\ yield = \frac{actual\ yield}{theoretical\ yield} \times 100\%}$$

- ✓ Actual and theoretical yields may be expressed as masses or molar amounts by any appropriate property such as mass, molar amount, volume, etc.

Example: The phosphorus pentoxide used to produce phosphoric acid for cola soft drinks is prepared by burning phosphorus in oxygen.

(a) What is the limiting reactant when 0.200 mol of P_4 and 0.200 mol of O_2 react according to

$$\mathbf{P_4 + 5O_2 \longrightarrow P_4O_{10}}$$

(b) Calculate the percent yield if 10.0 g of P_4O_{10} is isolated from the reaction.(molar mass of $P_4\ O_{10}$ = 283.88 g)

Solution:

a) According to the balanced chemical equation, 1 mol of P_4 reacts with 5 mol of O_2

- Hence, for 0.200 mol of P_4 , 1 mol of O_2 is required.
- ✓ But there is only 0.200 mol of O_2

Therefore, the limiting reactant is O_2

b) 5 mol of O_2 produces 1 mol of P_4O_{10}

- 0.200 mol of O_2 produces 0.040 mol of P_4O_{10}
- 1 mol of P_4O_{10} = 283.88 g
- Therefore 0.040 mol P_4O_{10} = 11.355 g (Theoretical yield)

Actual yield = 10.0 g

Therefore, $percentage\ yield = \frac{actual\ yield}{theoretical\ yield} \times 100\%$

$$= \frac{10.0g}{11.355\ g} \times 100\% = \mathbf{88.1\%}$$

4.5. Quantitative Chemical Analysis

Quantitative analysis: The determination of the concentration of a substance in a sample.

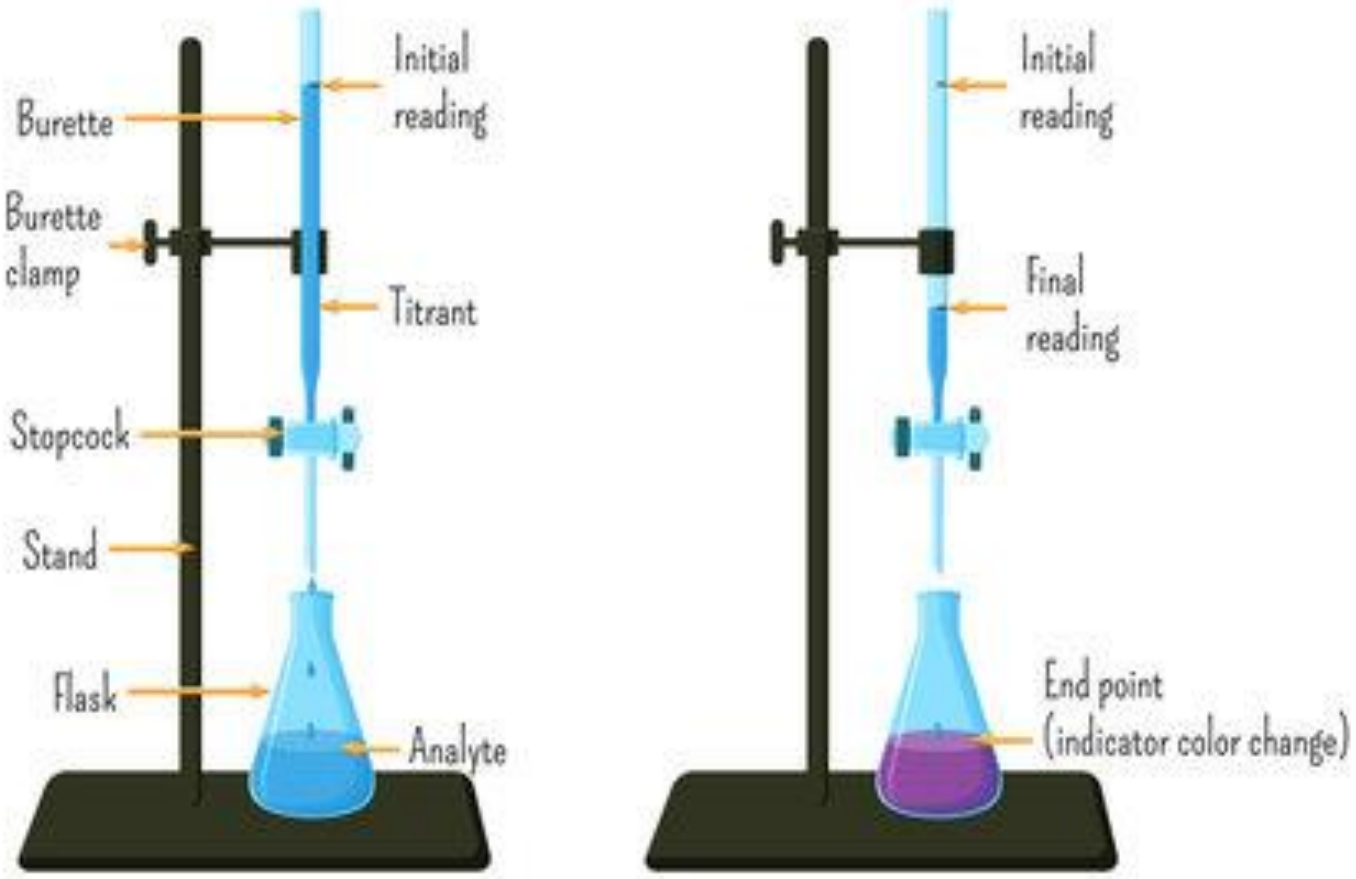
4.5.1. Acid-base Titration

Titration: A method of chemical analysis that involves incremental additions of a solution containing a known concentration of some substance (the **titrant**) to a sample solution containing the substance whose concentration is to be measured (the **analyte**).

A buret is used to make incremental additions of the analyte and to measure the volume used at the equivalence point

- ➢ The titrant and analyte undergo a chemical reaction of known stoichiometry
- ✓ Measuring the volume of titrant solution required for complete reaction with the analyte (the **equivalence point** of the titration) allows calculation of the analyte concentration

TITRATION


Initial reading
Burette
Burette clamp
Titrant
Stopcock
Stand
Flask
Analyte
Initial reading
Final reading
End point (indicator color change)

Equivalence points may be detected

i. **Using Indicators:** dyes that are added to the sample solutions to impart a change in color at or very near the equivalence point of the titration.

ii. by measuring some solution property that changes in a predictable way during the course of the titration.

E.g. **electrical property**

End point: The volume of titrant actually measured

Acid-base titration: The determination of the concentration of a sample of an acid solution by titrating it with a known concentration of a base.

Example: The end point in a titration of a 50.00 mL sample of aqueous HCl was reached by addition of 35.23 mL of 0.250 M NaOH titrant.

The titration reaction is:

$HCl(aq) + NaOH(aq) \longrightarrow NaCl(aq) + H_2O(l)$

What is the molarity of the HCl?

Solution:

According to the balanced equation, 1 mol of HCl reacts with 1 mol of NaOH.

Therefore,

M (HCl) × V (HCl) = M(NaOH) × V(NaOH)

$$M\ (HCl) = \frac{M(NaOH)}{V(HCl)} \times V(NaOH) = \frac{0.250M}{50.00mL} \times 35.23mL = \mathbf{0.176\ M}$$

4.5.2. Gravimetric Analysis

Gravimetric analysis: One in which a sample is subjected to some treatment that causes a change in the physical state of the analyte that permits its separation from the other components of the sample.

- The required change of state in a gravimetric analysis may be achieved by various physical and chemical processes.

Examples:

- The moisture content of a sample is determined by measuring the mass of a sample before and after it is subjected to a controlled heating process that evaporates the water.
- The analyte is subjected to a precipitation reaction and the precipitate is isolated from the reaction mixture by filtration.

Example: What is the percent of chloride ion in a sample if 1.1324 g of the sample produces 1.0881 g of AgCl when treated with excess Ag^+?

$$\mathbf{Ag^+_{(aq)} + Cl^-_{(aq)} \longrightarrow AgCl_{(s)}}$$

(molar masses: Cl^- = 35.45 g = AgCl = 143.32 g)

Solution: 143.32 g AgCl contains 35.45 g Cl^{-1}

1.0881 g AgCl contains $\frac{35.45g}{143.32g} \times 1.0881g = 0.2691\ g$

If 0.2691 g is obtained from 1.1324 g sample, its mass percent is

$$\frac{0.2691}{1.1324} \times 100\% = \mathbf{23.76\%}$$